

Propofol: Drug information - UpToDate

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For additional information see "Propofol: Patient drug information" and "Propofol: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Diprivan;
- Diprivan +RFID

Brand Names: Canada

- Diprivan;
- TEVA-Propofol [DSC]

Pharmacologic Category

- General Anesthetic

Dosing: Adult

Dosage guidance:

Safety: Consult local/state regulations and institutional policies and procedures regarding administration and monitoring requirements prior to use; dosing practices may vary.

General anesthesia

General anesthesia:

Note: Individualize dosing based on patient factors and concomitant anesthetic agents.

Induction of anesthesia: Note: In high-risk patients (eg, debilitated, American Society of Anesthesiologists - Physical Status [ASA-PS] 3 or 4, hemodynamically compromised), avoid rapid bolus administration to minimize hypotension. Induction doses may be administered in divided doses.

Healthy adults, ASA-PS 1 or 2: **IV:** Usual total dose: 1 to 2.5 mg/kg.

Hemodynamic compromise or hypovolemia: **IV:** Usual total dose: 0.5 to 1.5 mg/kg.

Debilitated or ASA-PS 3 or 4: **IV:** Usual total dose: 0.5 to 1.5 mg/kg.

Maintenance of anesthesia:

Healthy adults, ASA-PS 1 or 2:

Continuous IV infusion: Usual dosing range: 50 to 200 mcg/kg/minute; titrate to clinical response.

Debilitated or ASA-PS 3 or 4:

Continuous IV infusion: Usual dosing range: 50 to 100 mcg/kg/minute; titrate to clinical response.

Rapid sequence intubation outside the operating room

Rapid sequence intubation outside the operating room (induction): Note: May cause dose-related hypotension; consider alternative agent in patients who are hemodynamically unstable (Ref). Refer to local/state regulations and institutional policies and procedures regarding administration and monitoring requirements.

IV: 1.5 to 2 mg/kg once; usual dose range: 1 to 3 mg/kg. Some experts decrease dose in the setting of hemodynamic instability (eg, ~50% dose reduction **or** administer dose in 10 mg increments until patient is unconscious) (Ref). **Note:** Continuous infusion may be started if longer sedation is required (Ref).

Sedation

Sedation:

Mechanically ventilated patients in the ICU, sedation: Note: Used as part of a multimodal strategy (eg, combination of sedatives and analgesics) for ICU sedation and preferred over a benzodiazepine due to less risk of prolonged sedation and improved time to extubation; titrate to maintain a light level of sedation (eg, Richmond Agitation Sedation Scale 0 to -2) or clinical effect (eg, ventilator synchrony) (Ref).

Continuous IV infusion: Initial: 5 mcg/kg/minute; increase by 5 to 10 mcg/kg/minute every 5 to 10 minutes until goal sedation level is achieved. Usual maintenance dose: 5 to 50 mcg/kg/minute. Maximum dose (not well defined; may vary by institution): 60 to 80 mcg/kg/minute. Daily awakening trials or use of nurse-protocolized sedation are recommended; generally, titrate down slowly to avoid rapid awakening. If agitated after discontinuation of continuous infusion, then restart at ~50% of the previous maintenance dose (Ref).

Monitored anesthesia care: Note: Individualize dosing based on patient factors and concomitant anesthetic agents.

Healthy adults, ASA-PS 1 or 2:

Continuous IV infusion: Initial: 25 to 75 mcg/kg/minute; titrate to adequate sedation. **Note:** If rapid effect is desired, may initiate with a higher infusion rate (eg, 100 to 150 mcg/kg/minute for 3 to 5 minutes) or administer an initial bolus (eg, 0.25 to 0.5 mg/kg) (Ref).

or

IV: Intermittent bolus: 10 to 20 mg; may give additional doses as needed to achieve adequate sedation (Ref).

Debilitated or ASA-PS 3 or 4: Use reduced dose. Bolus slowly and avoid rapid repeat dosing.

Procedural sedation, outside the operating room (off-label use): Note: Monitor respiratory and cardiovascular systems; refer to local/state regulations and institutional policies and procedures regarding administration and monitoring requirements.

IV: Initial: 0.5 to 1 mg/kg, followed by 0.25 to 0.5 mg/kg every 1 to 3 minutes, as needed, to achieve adequate sedation (Ref).

Status epilepticus, refractory

Status epilepticus, refractory (off-label use):

Note: Used as an alternative or adjunctive agent to midazolam, barbiturates, or ketamine after conventional intermittent therapies have failed. Mechanical ventilation and hemodynamic support required; continuous EEG is recommended. Titrate doses to cessation of electrographic seizures or burst suppression (Ref). Optimal regimen and dose are uncertain; refer to institutional protocol.

IV: Loading dose: 1 to 2 mg/kg, followed by 0.5 to 2 mg/kg every 3 to 5 minutes until seizures are suppressed; maximum total dose: 10 mg/kg (Ref).

Continuous IV infusion: After initial loading dose, begin continuous infusion at an initial rate of 20 mcg/kg/minute; titrate to cessation of electrographic seizures or burst suppression. Usual dose range: 30 to 60 mcg/kg/minute. Maximum dose (not well defined and may vary by institution): 200 mcg/kg/minute. Use caution with doses >80 mcg/kg/minute for >48 hours (Ref).

For breakthrough status epilepticus, administer bolus of 0.5 to 2 mg/kg every 3 to 5 minutes in addition to increasing continuous infusion rate by 5 to 10 mcg/kg/minute every 5 minutes (Ref).

Note: Generally, a period of at least 24 hours of electrographic suppression is suggested prior to down titrating the continuous infusion; withdraw gradually by decreasing the dose 15% to 20% every 3 hours while continuing conventional therapies (Ref).

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: No dosage adjustment necessary for any degree of kidney dysfunction (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be dialyzed (large V_d , highly protein bound): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Unlikely to be dialyzed (large V_d , highly protein bound): No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

Note: Case reports of filter clotting in patients with propofol-associated hypertriglyceridemia receiving CRRT have been reported; monitor triglycerides and visually inspect the CRRT system for evidence of congealing or a “milky brown” appearance of the “clot” as a potential cause (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Note: Similar concerns for a potential for filter clotting exist in patients receiving PIRRT who experience propofol-associated hypertriglyceridemia; monitor triglycerides and visually inspect the PIRRT system for evidence of congealing or a “milky brown” appearance of the “clot” as a potential cause (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Hepatic Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Despite extensive hepatic metabolism, the pharmacokinetics of propofol in the setting of cirrhosis were not significantly altered compared to healthy controls (Ref).

Hepatic impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C: No dosage adjustment necessary (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥ 30 kg/m²):

IV: Use **adjusted body weight** for initial weight-based dose calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, adjusted body weight to/from actual body weight) (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Weight-based dosing calculations using actual body weight for bolus doses or continuous infusions may lead to excessive concentrations (Ref); therefore, use of an alternative size descriptor (eg, adjusted body weight) is recommended for dosing propofol in patients with obesity (Ref). The majority of studies evaluated pharmacokinetic and pharmacodynamics of propofol when used during induction and/or maintenance of anesthesia (Ref). Even though lean body weight (LBW) appears to have the best correlation to volume of distribution and clearance, the use of the complex LBW formula is not recommended because of concerns with calculation errors, especially in critical situations, and concerns with underdosing in situations where rapid sedation is preferred (eg, induction of anesthesia) (Ref).

Dosing: Older Adult

Refer to adult dosing specific indications, “Debilitated or ASA-PS 3 or 4.” Smaller doses are generally required in older adults and when used with opioids. Dosage must be individualized and titrated to the desired clinical effect. Wait at least 3 to 5 minutes between dosage adjustments to clinically assess drug effects. Prior to use, consult local/state regulations and institutional policies and procedures regarding administration and monitoring requirements.

Dosing: Pediatric

(For additional information see “Propofol: Pediatric drug information”)

Dosage guidance:

Safety: Consult local regulations and individual institutional policies and procedures; should only be used by experienced personnel who are not actively engaged in the procedure or surgery. If used in a nonintubated and/or nonmechanically ventilated patient, qualified personnel and appropriate equipment for rapid institution of respiratory and/or cardiovascular support must be immediately available.

Dosing: Dosage must be individualized based on total body weight and titrated to the desired clinical effect; the following are general dosing guidelines.

Anesthesia, general

Anesthesia, general : Note: Individualize dosing based on patient factors and concomitant anesthetic agents.

Induction of general anesthesia: Children ≥ 3 years and Adolescents < 17 years: IV: 2.5 to 3.5 mg/kg over 20 to 30 seconds. **Note:** Use a lower dose for ASA-PS III or IV.

Maintenance of general anesthesia: Infants ≥ 2 months, Children, and Adolescents:

Intermittent IV bolus: Initial: IV: 1 to 4 mg/kg once; may administer additional smaller bolus doses if clinical signs of light anesthesia present; usual bolus dose: 0.5 to 2 mg/kg/dose; repeat as needed based on clinical response and desired level of anesthesia.

Continuous IV infusion: Initial dose immediately following induction: 200 to 300 mcg/kg/minute (12 to 18 mg/kg/hour); then decrease dose after 30 minutes if clinical signs of light anesthesia are absent; usual infusion rate after initial 30 minutes: 125 to 150 mcg/kg/minute (7.5 to 9 mg/kg/hour). **Note:** Younger pediatric patients may need higher infusion rates than older pediatric patients.

ICU sedation

ICU sedation (critically ill): Limited data available: **Note:** Prior to use, consult local/state regulations and institutional policies and procedures regarding administration and monitoring requirements; dosing practices may vary.

Infants, Children, and Adolescents:

Loading dose: IV: 0.5 to 1 mg/kg once; maximum dose: 50 mg/dose (Ref).

Continuous IV infusion: Usual range: 16 to 66 mcg/kg/minute (1 to 4 mg/kg/**hour**) (Ref). **Note:** Monitor closely for signs and symptoms of propofol-related infusion syndrome (PRIS).

Procedural sedation

Procedural sedation: Limited data available:

Infants, Children, and Adolescents:

Repeated IV bolus method: IV: Initial dose: 1 to 2 mg/kg; higher initial doses of 2 mg/kg are recommended in infants and children <3 years of age; follow initial dose with 0.5 to 1 mg/kg every 3 to 5 minutes as needed until adequate level of sedation achieved (Ref).

IV bolus followed by continuous IV infusion: IV: Initial bolus: 1 to 2 mg/kg followed by continuous IV infusion: Dose range: 50 to 250 mcg/kg/minute (3 to 15 mg/kg/**hour**); titrate to desired level of sedation (Ref).

Propofol with concurrent ketamine; emergency department procedures: IV: 0.5 to 0.75 mg/kg (Ref).

Status epilepticus; refractory

Status epilepticus; refractory: Limited data available:

Children and Adolescents:

Initial propofol infusion: IV: Loading dose 1 to 2 mg/kg once, then initiate continuous IV infusion at 20 mcg/kg/minute (1.2 mg/kg/**hour**); titrate to desired effect (eg, burst suppression on EEG); usual dose range: 30 to 200 mcg/kg/minute (1.8 to 12 mg/kg/**hour**) (Ref). **Note:** Use caution when administering high doses (>65 mcg/kg/minute [**>3.9 mg/kg/hour**]) for extended periods of time (>48 hours); monitor closely for adverse effects (eg, propofol-related infusion syndrome [PRIS]); Neurocritical Care Society guidelines for status epilepticus state that use of propofol in young children is contraindicated (Ref).

Breakthrough seizure while on propofol infusion: IV: Increase continuous IV infusion rate by 5 to 10 mcg/kg/minute (0.3 to 0.6 mg/kg/**hour**) every 5 minutes with or without an additional 1 mg/kg IV bolus (Ref).

Dosing: Kidney Impairment: Pediatric

There are no pediatric-specific recommendations in the manufacturer's labeling. In adults, no dosage adjustments are recommended.

Dosing: Liver Impairment: Pediatric

There are no pediatric-specific recommendations in the manufacturer's labeling. In adults, no dosage adjustments are recommended.

Adverse Reactions (Significant): Considerations

Cardiac conduction disturbances

Propofol may cause bradyarrhythmias (**bradycardia**) and convert tachyarrhythmias (**tachycardia**) to sinus rhythm (Ref). May also be associated rarely with **prolonged QT interval on ECG** (Ref) and **ECG abnormality** (shortening of the QT interval) (Ref).

Mechanism: Not clearly established; may be direct or indirect (more emerging data). Indirect effect may be related to modulation of autonomic nervous system tone through changes in the baroreceptor reflex, resulting in bradycardia (Ref). QT prolongation is most likely relegated to dose and predisposed cardiac conditions (Ref).

Onset: Rapid; both bradycardia and QT prolongation have been reported to occur rapidly (Ref).

Risk factors:

Bradycardia (possible risk factors):

- High doses, prolonged duration of therapy (in association with propofol-related infusion syndrome [PRIS]) (Ref)
- Concurrent medications, such as beta-blockers, neostigmine, neuromuscular blockers, opioids (Ref)
- Previous cardiac history (Ref)
- Surgical issues, such as extradural anesthesia, previous syncope, preoperative conduction abnormalities, too light anesthesia (Ref)

QT prolongation:

- Drug-specific:
 - High doses (Ref)
 - Pediatric patients (in association with PRIS) (Ref)
- General:
 - Females (Ref)
 - Age >65 years (Ref)
 - Structural heart disease (eg, history of myocardial infarction or heart failure with reduced ejection fraction) (Ref)
 - Genetic defects of cardiac ion channels (Ref)
 - History of drug-induced torsades de pointes (Ref)
 - Congenital long QT syndrome (Ref)
 - Baseline QTc interval prolongation (eg, >500 msec) or lengthening of the QTc by ≥ 60 msec (Ref)
 - Electrolyte disturbances (eg, hypocalcemia, hypokalemia, hypomagnesemia) (Ref)
 - Bradycardia (Ref)
 - Hepatic impairment (Ref)
 - Kidney impairment (Ref)
 - Loop diuretic use (Ref)
 - Sepsis (Ref)
 - Concurrent administration of multiple medications (≥ 2) that prolong the QT interval or medications with drug interactions that increase serum concentrations of QT-prolonging medications (Ref)

Hypersensitivity reactions (immediate)

Propofol may rarely cause **anaphylaxis** (Ref), **angioedema**, and **bronchoconstriction** (Ref). Current evidence suggests that immediate hypersensitivity reactions associated with propofol are rare compared to other medications often administered at the same time (eg, antibiotics, neuromuscular blocking agents).

Mechanism: Non-dose-related; immunologic or nonimmunologic. Immediate hypersensitivity reactions (eg, anaphylaxis, urticaria) can be IgE-mediated or non-IgE-mediated (Ref). Propofol allergy may be due to the presence of the diisopropyl side chain or phenol group (Ref).

Onset: Immediate hypersensitivity reactions: rapid; occur shortly after administration (often within 5 to 10 minutes) (Ref); reported to occur after first exposure (Ref) or after multiple exposures to propofol (Ref).

Risk factors:

- Food allergy: Although propofol is mixed in a formulation containing soybean oil and egg lecithin, true allergy has been associated with soy and egg proteins and not these oils. Current evidence suggests that immediate hypersensitivity reactions associated with propofol are not predicted by a

history of food allergy, including those with soy, egg, and peanut allergy (Ref). Available studies have indicated that propofol may be used safely in soy- or egg-allergic patients (Ref). A statement issued by the American Academy of Allergy, Asthma and Immunology has concluded that patients with soy or egg allergy can receive propofol without any special precautions (Ref).

Hypertriglyceridemia

Hypertriglyceridemia can occur with propofol, which may lead to acute **pancreatitis**. Rare cases of propofol-induced acute pancreatitis have occurred in the absence of hypertriglyceridemia (Ref).

Mechanism: Dose-and duration-related; propofol is formulated as a ~10% lipid emulsion, which can cause hypertriglyceridemia and lead to acute pancreatitis (Ref). Cases of propofol-induced acute pancreatitis that occur in the absence of hypertriglyceridemia have been described as idiosyncratic (Ref).

Onset: Rapid; hypertriglyceridemia may occur at any time during therapy, but the median time ranges from 2 to 4 days (Ref).

Risk factors:

Hypertriglyceridemia:

- Dose and duration: >50 mcg/kg/minute for >2 days (Ref)
- Higher severity of illness (Ref)
- Longer ICU stay (Ref)
- SARS-CoV-2 positive (Ref)
- Younger patients (Ref)

Acute pancreatitis:

- Preexisting diseases or factors that predispose to acute pancreatitis (eg, alcohol use disorder, abdominal or cardiothoracic surgeries, hypertriglyceridemia, diabetes, family history of lipoproteinemia, obesity) (Ref)

Hypotension

Propofol may cause severe **hypotension** (to the effect of $\geq 30\%$ decrease in mean arterial pressure [MAP]), especially with bolus dosing or in the setting of hypovolemia, sepsis, or shock (Ref). May also induce hypotension in pediatric patients (neonates) and trauma patients (adults and pediatrics) (Ref). Propofol in neonatal endotracheal intubations may potentiate hypotension, leading to serious complications, such as intraventricular hemorrhage or periventricular leukomalacia (Ref).

Mechanism: Not clearly established; proposed mechanisms include reduction of peripheral resistance through a direct action in vascular smooth muscle (possibly from histamine release), inhibition of sympathetic activity on the vasculature (Ref), and reduction in stressed volume caused by decreased venous and arterial resistance with no change in cardiac output (Ref). Propofol acts as a calcium channel blocker and beta-adrenergic antagonist (Ref).

Onset: Rapid; propofol may induce hypotension after bolus administration, new continuous infusion initiation, or increases in the dose.

Risk factors:

- Bolus dosing
- Hypovolemia
- Older, debilitated, or American Society of Anesthesiologists - Physical Status (ASA-P) 3 or 4 patients
- Baseline MAP 60 to 70 mm Hg prior to propofol administration (Ref)
- Changes to propofol infusion rate (Ref)
- Need for renal replacement therapy during treatment with propofol (Ref)

- Rapid sequence intubation in patients (adults and pediatrics) with trauma (Ref)
- Sepsis (adults) (Ref)
- Severe aortic stenosis (Ref)
- Trauma patients (adults): Age >55 years, lower baseline systolic blood pressure, obesity (Ref)
- Preterm infants 29 to 32 gestational weeks with infant respiratory distress syndrome (Ref)

Propofol-related infusion syndrome (PRIS)

Propofol-related infusion syndrome (PRIS), is a serious adverse reaction that has been reported in adult and pediatric patients (Ref). PRIS is rare but has a high mortality rate and is characterized by a constellation of symptoms, including dysrhythmia (**bradycardia** or **tachycardia**), widening of the QRS complex, heart failure, **hypotension**, **asystole**, lipemia and **hypertriglyceridemia**, **metabolic acidosis**, and/or **rhabdomyolysis** or myoglobinuria with acute kidney injury and hyperkalemia (Ref).

Mechanism: Dose- and duration-related; may be due to a mitochondrial defect (Ref). It is thought that propofol inhibits transportation of long-chain fatty acids into the cell and interferes with/uncouples oxidative phosphorylation (Ref).

Onset: Rapid; between 1 to 4 days (Ref).

Risk factors:

- Dose and duration: Usually ≥ 4 mg/kg/hour for ≥ 2 days (Ref), but has also been reported following large dose, short-term infusions during surgical anesthesia. PRIS has also been reported with lower-dose infusions (Ref)
- Young age (Ref)
- Catecholamine, corticosteroid, or vasopressor use (Ref)
- Critical illness (Ref)
- High fat, low carbohydrate intake (Ref)
- Inborn error of fatty acid oxidation (Ref)
- Males (Ref)
- Decreased oxygen delivery to tissues
- Sepsis
- Serious neurological injury

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified. Reported adverse reactions are for adults and pediatrics.

Frequency not defined:

Cardiovascular: Acute myocardial infarction, asystole, atrial fibrillation, atrial premature contractions, atrioventricular block, bigeminy, bundle branch block, chest pain, depression of ST segment on ECG, edema, extrasystoles, flushing, hypertension, ischemic heart disease, low cardiac output, right heart failure, supraventricular tachycardia, syncope, tachycardia, ventricular fibrillation, ventricular premature contractions, ventricular tachycardia

Dermatologic: Diaphoresis, pruritus, skin rash, urticaria

Endocrine & metabolic: Dehydration, hyperglycemia, hyperkalemia, hyperlipidemia, hypomagnesemia, respiratory acidosis (during weaning), serum hyperosmolality

Gastrointestinal: Abdominal cramps, diarrhea, hiccups, intestinal obstruction, nausea, parotid gland enlargement, sialorrhea, sore throat (burning), vomiting, xerostomia

Genitourinary: Oliguria, urinary retention

Hematologic & oncologic: Disorder of hemostatic components of blood, hemorrhage, leukocytosis

Hepatic: Abnormal liver function

Infection: Sepsis

Local: Burning sensation at injection site, erythema at injection site, injection-site phlebitis, injection-site pruritus, itching at injection site, pain at injection site (includes stinging), skin discoloration at injection site, urticaria at injection site

Nervous system: Abnormal dreams, abnormality in thinking, anticholinergic syndrome, anxiety, asthenia, atypical sexual behavior, awareness under anesthesia without pain, chills, confusion, delirium, depression, dizziness, drowsiness, emotional lability, euphoria, fatigue, hallucination, headache, hypertonia, hypotonia, hysteria, insomnia, intracranial hypertension, neuropathy, pain, paresthesia, twitching

Neuromuscular & skeletal: Laryngospasm, limb pain, muscle rigidity, myalgia, neck stiffness

Ophthalmic: Amblyopia, conjunctival hyperemia, diplopia, eye pain, nystagmus disorder, visual disturbance

Otic: Otagia, tinnitus

Renal: Increased blood urea nitrogen, increased serum creatinine, renal failure syndrome

Respiratory: Airway obstruction, apnea, bronchospasm, cough, decreased lung function, dyspnea, hyperventilation, hypoventilation, pharyngitis, sneezing, tachypnea, wheezing

Miscellaneous: Fever

Postmarketing:

Cardiovascular: Bradycardia (Baumgarten 2021), ECG abnormality (shortening of the QT interval) (Erdil 2009), hypotension (Baumgarten 2021), phlebitis, prolonged QT interval on ECG (Kim 2008)

Endocrine & metabolic: Hypertriglyceridemia (Corrado 2020), metabolic acidosis (Chukwuemeka 2006)

Gastrointestinal: Dysgeusia (Farzana 2022), pancreatitis (Asghar 2020)

Genitourinary: Cloudy urine (including urate crystalluria) (Baumgarten 2021), urine discoloration (green) (Baumgarten 2021)

Hypersensitivity: Anaphylaxis (Koul 2011), angioedema (You 2012), infusion-related reaction (propofol-related infusion syndrome) (Krajčová 2015), nonimmune anaphylaxis (McHale 1992)

Local: Venous thrombosis at injection site

Nervous system: Altered sense of smell (Farzana 2022), anosmia (Farzana 2022), myoclonus (including perioperative; including seizure and opisthotonos) (Tam 2009), neurotoxicity (neuroexcitatory reaction [propofol “frenzy”]; including abnormal eye movements, agitation, clonus, dystonia, impaired consciousness, involuntary body movements [bucking, jerking, and thrashing], shivering, tremor) (Panchamia 2022)

Neuromuscular & skeletal: Rhabdomyolysis (Zarovnaya 2007)

Respiratory: Bronchoconstriction (Nishiyama 2001), hypoxia (Hanna 1998), pulmonary edema (Waheed 2014)

Contraindications

Hypersensitivity to propofol or any component of the formulation; hypersensitivity to eggs, egg products, soybeans, or soy products.

Note: Although the manufacturer’s labeling lists egg allergy as a contraindication, available studies (mostly retrospective) and an American Academy of Allergy, Asthma, and Immunology statement

have suggested that propofol may be used safely in soy- or egg-allergic patients (AAAAI [Lieberman 2015]; AAAAI 2019; Asserhøj 2016; Dziedzic 2016; Murphy 2011). In patients with more severe soy or egg allergy, consider the use of an alternative anesthetic or a small trial dose of propofol prior to full dose administration (Johnson 2021).

Canadian labeling: Additional contraindication (not in US labeling): Hypersensitivity to lipid emulsions; sedation of children ≤ 18 years of age receiving intensive care.

Warnings/Precautions

Concerns related to adverse effects:

- Injection-site reaction: Transient local pain may occur during IV injection; lidocaine 1% solution may be administered prior to administration or may be added to propofol immediately prior to administration to reduce pain associated with injection.

Disease-related concerns:

- Increased intracranial pressure: Use with caution in patients with increased intracranial pressure or impaired cerebral circulation; propofol may cause substantial decreases in mean arterial pressure with subsequent decreases in cerebral perfusion pressure; to minimize risk, consider continuous infusion or administer as a slow bolus.
- Infection risk: Propofol vials and prefilled syringes have the potential to support the growth of various microorganisms despite product additives intended to suppress microbial growth. To limit the potential for contamination, strictly adhere to recommendations in product labeling for handling and administering propofol.
- Respiratory disease: Use with caution in patients with respiratory disease.
- Seizure disorder: Use with caution in patients with a history of epilepsy or seizures; seizure may occur during recovery phase.

Concurrent drug therapy issues:

- Opioids: Concomitant use may lead to increased sedative or anesthetic effects of propofol, more pronounced decreases in systolic, diastolic, and mean arterial pressures and cardiac output; lower doses of propofol may be needed. In addition, fentanyl may cause serious bradycardia when used with propofol in pediatric patients. Alfentanil use with propofol has precipitated seizure activity in patients without any history of epilepsy.

Special populations:

- Pediatric neurotoxicity: In pediatric and neonatal patients < 3 years of age and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems including neurodevelopmental delay (and related diagnoses), learning disabilities, and ADHD. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.

- **Edetate disodium:** Some formulations contain edetate disodium which may lead to decreased zinc levels in patients with prolonged therapy (>5 days) or a predisposition to zinc deficiency (eg, burns, diarrhea, or sepsis). A holiday from propofol infusion should take place after 5 days of therapy to allow for evaluation and necessary replacement of zinc.
- **Sulfites:** Some formulations may contain sulfites.

Other warnings/precautions:

- **Abrupt discontinuation:** Avoid abrupt discontinuation prior to weaning or daily wake up assessments. Abrupt discontinuation can result in rapid awakening, anxiety, agitation, and resistance to mechanical ventilation; wean the infusion rate so the patient awakens slowly. Discontinue opioids and paralytic agents prior to weaning. Long-term infusions can result in some tolerance; taper propofol infusions to prevent withdrawal.
- **Analgesic supplementation:** Propofol lacks analgesic properties; pain management requires specific use of analgesic agents, at effective dosages, propofol must be titrated separately from the analgesic agent.
- **Experienced personnel:** Use requires careful patient monitoring, should only be used by experienced personnel who are not actively engaged in the procedure or surgery. If used in a nonintubated and/or nonmechanically ventilated patient, qualified personnel and appropriate equipment for rapid institution of respiratory and/or cardiovascular support must be immediately available. Use to induce moderate (conscious) sedation in patients warrants monitoring equivalent to that seen with deep anesthesia. Consult local regulations and individual institutional policies and procedures.

Warnings: Additional Pediatric Considerations

In pediatric and neonatal patients <3 years of age and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on the child's or fetus' brain development and may contribute to various cognitive and behavioral problems; the FDA is requiring warnings be included in the manufacturer's labeling for all general anesthetic/sedative drugs. Multiple animal species studies have shown adverse effects on brain maturation; in juvenile animals, drugs that potentiate GABA activity and/or block NMDA receptors for >3 hours demonstrated widespread neuronal and oligodendrocyte cell loss along with alteration in synaptic morphology and neurogenesis. Epidemiological studies in humans have reported various cognitive and behavioral problems including neurodevelopmental delay (and related diagnoses), learning disabilities, and ADHD. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. Further studies are needed to fully characterize findings and ensure that these findings are not related to underlying conditions or the procedure itself. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Use extreme caution when using for refractory status epilepticus because higher doses are often required to control seizures for an extended period of time (in pediatric patients: doses >65 mcg/kg/minute [4 mg/kg/hour] for >48 hours) which may increase risk for propofol-related infusion syndrome (NCS [Brophy 2012]). Fatal cardiac failure has also been reported with concurrent ketogenic diet and propofol therapy (Baumeister 2004).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Emulsion, Intravenous:

Diprivan: 100 mg/10 mL (10 mL); 200 mg/20 mL (20 mL); 500 mg/50 mL (50 mL); 1000 mg/100 mL (100 mL) [contains edetate (edta) disodium, egg phospholipids (egg lecithin), glycerin, soybean oil]

Generic: 1000 mg/100 mL (100 mL); 200 mg/20 mL (20 mL); 500 mg/50 mL (50 mL)

Emulsion, Intravenous [preservative free]:

Diprivan +RFID: 200 mg/20 mL (20 mL) [contains edetate (edta) disodium, egg phospholipids (egg lecithin), soybean oil]

Generic: 1000 mg/100 mL (100 mL); 200 mg/20 mL (20 mL); 500 mg/50 mL (50 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Emulsion (Diprivan +RFID Intravenous)

200 mg/20 mL (per mL): \$0.37

Emulsion (Diprivan Intravenous)

100 mg/10 mL (per mL): \$0.14

200 mg/20 mL (per mL): \$0.14

500 mg/50 mL (per mL): \$0.14

1000 mg/100 mL (per mL): \$0.14

Emulsion (Propofol Intravenous)

200 mg/20 mL (per mL): \$0.13 - \$0.36

500 mg/50 mL (per mL): \$0.13 - \$0.36

1000 mg/100 mL (per mL): \$0.12 - \$0.36

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Emulsion, Intravenous:

Diprivan: 10 mg/mL (20 mL, 50 mL, 100 mL) [contains edetate (edta) disodium, egg phosphatides, soybean oil]

Generic: 10 mg/mL (20 mL, 50 mL, 100 mL)

Administration: Adult

IV: Administer intermittent bolus or by continuous IV infusion. Prior to use, consult local/state regulations and institutional policies and procedures regarding administration and monitoring requirements. Strict aseptic technique must be maintained in handling although a preservative has been added. Do not use if contamination is suspected. Do not administer through the same IV catheter with blood or plasma. Tubing and any unused portions of propofol vials should be discarded after 12 hours.

To reduce pain associated with injection, use larger veins of forearm or antecubital fossa; lidocaine IV (1 mL of a 1% solution) may also be used prior to administration or it may be added to propofol immediately before administration in a quantity not to exceed 20 mg lidocaine per 200 mg propofol. Do not use filter <5 micron for administration.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry cold or dry warm compresses. Initiate hyaluronidase antidote for refractory cases in addition to supportive management (Ref).

Hyaluronidase (refractory management): **Intradermal or SUBQ:** Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if infiltrated catheter remains in place, administer through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Preparation for Administration: Adult

Does not need to be diluted; however, propofol may be further diluted in 5% dextrose in water to a concentration of ≥ 2 mg/mL. Diluted propofol solutions are more stable in glass compared to plastic (95% potency after 2 hours when stored in plastic).

Administration: Pediatric

Note: Consult local regulations and individual institutional policies and procedures.

Parenteral: IV: Propofol 1% (10 mg/mL) (eg, Diprivan): Strict aseptic technique must be maintained in handling. Shake emulsion well before use. May be administered undiluted or may be further diluted with D5W. Do not administer via filter with <5 -micron pore size. Do not administer through the same IV catheter with blood or plasma. Tubing and any unused portions of propofol vials should be discarded after 12 hours.

To reduce pain associated with bolus injection, use larger veins of forearm or antecubital fossa; lidocaine IV (1 mL of a 1% solution) may also be used prior to administration or lidocaine may be added to propofol immediately before administration.

Induction: Administer bolus doses over 20 to 30 seconds.

Maintenance: Administer at a concentration of 2 to 10 mg/mL as intermittent bolus injection at prescribed rate or as a continuous IV infusion via an infusion pump.

Preparation for Administration: Pediatric

Parenteral: IV: May be further diluted in D5W to a concentration of ≥ 2 mg/mL; diluted emulsion is more stable in glass (stability in plastic: 95% potency after 2 hours). ASHP recommends a standard concentration of 10 mg/mL in pediatric patients. To reduce pain associated with administration, may add lidocaine to propofol immediately before administration in a quantity not to exceed 20 mg lidocaine per 200 mg propofol. Strict aseptic technique must be maintained in handling. Vials are intended for single-patient use and single-dose use only; discard remaining product. Shake emulsion well before use.

Vesicant/Extravasation Risk

Vesicant.

Storage/Stability

Store between 4°C to 25°C (40°F to 77°F); refrigeration is not required. Do not freeze. Shake well before use. Withdraw from vial into a syringe immediately after sterile vented spike inserted. Administration should begin immediately and completed within 12 hours after the vial has been opened. Do not use if there is evidence of excessive creaming or aggregation, large visible droplets, or separation of phases of emulsion.

Use: Labeled Indications

General anesthesia: Induction of anesthesia in patients ≥ 3 years of age; maintenance of anesthesia in patients ≥ 2 months of age.

Mechanically ventilated patients in the ICU, sedation: Sedation of intubated, mechanically ventilated adults in the ICU.

Monitored anesthesia care sedation: Initiation and maintenance of monitored anesthesia care sedation in adults.

Sedation and regional anesthesia: Combined sedation and regional anesthesia in adults.

Note: Consult local regulations and individual institutional policies and procedures.

Use: Off-Label: Adult

Procedural sedation, outside the operating room; Status epilepticus, refractory

Medication Safety Issues

Sound-alike/look-alike issues:

Diprivan may be confused with Diflucan, Ditropan

Propofol may be confused with fospropofol

Propofol may be confused with bupivacaine (liposomal) due to similar white, milky appearance.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Pediatric patients: High-risk medication:

KIDs List: Propofol is identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list; use should be avoided at doses >4 mg/kg/hour for >48 hours in neonatal and pediatric patients ≤18 years of age due to risk of propofol-related infusion syndrome (strong recommendation; moderate quality of evidence) (PPA [McPherson 2025]).

Administration issues:

Propofol may be confused with bupivacaine liposome injectable suspension (Exparel) in operating rooms and other surgical areas due to their similar white, milky appearance especially when prepared in syringes. Bupivacaine liposome injectable suspension (Exparel) is intended only for administration via infiltration into the surgical site (and **not** for systemic use). Confusion with propofol may lead to accidental intravenous administration of Exparel instead of the intended propofol. Therefore, to avoid potential confusion ISMP recommends that all vials be separated when stocked in common areas and all prepared syringes be labeled.

Metabolism/Transport Effects

Substrate of CYP1A2 (Minor), CYP2A6 (Minor), CYP2B6 (Minor), CYP2C19 (Minor), CYP2C9 (Minor), CYP2D6 (Minor), CYP2E1 (Minor), CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential; **Inhibits** CYP3A4 (Weak)

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alfentanil: May increase adverse/toxic effects of Propofol. Specifically the development of opisthotonus (severe hyperextension and spasticity resulting in arching or bridging position) and/or tonic clonic seizures. *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

ALPRAZolam: CYP3A4 Inhibitors (Weak) may increase serum concentrations of ALPRAZolam. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amisulpride (Oral): May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Azithromycin (Systemic): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Azithromycin (Systemic). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bucizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cannabis: May decrease therapeutic effects of Propofol. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CarBAMazepine: CYP3A4 Inhibitors (Weak) may increase serum concentrations of CarBAMazepine. *Risk C: Monitor*

Carbetocin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Carbetocin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Ceritinib: Propofol may increase bradycardic effects of Ceritinib. Propofol may increase QTc-prolonging effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. Also monitor for QTc prolongation and ventricular arrhythmia. *Risk D: Consider Therapy Modification*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chloroquine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Chloroquine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Clofazimine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Clofazimine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Propofol. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dabrafenib: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diphenhydramine (Systemic): CNS depressant effects of CNS Depressants and Diphenhydramine (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Domperidone: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

Droperidol: May increase CNS depressant effects of Propofol. Propofol may increase QTc-prolonging effects of Droperidol. Management: Consider dose reductions of droperidol or propofol during coadministration due to additive effects on CNS depression. Also monitor for QTc prolongation and ventricular arrhythmia. *Risk D: Consider Therapy Modification*

Duloxetine: Blood Pressure Lowering Agents may increase hypotensive effects of Duloxetine. *Risk C: Monitor*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Encorafenib: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these

agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

EPHEDrine (Systemic): Propofol may increase therapeutic effects of EPHEDrine (Systemic). *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Etelcalcetide: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Etelcalcetide. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

FentaNYL: Propofol may increase CNS depressant effects of FentaNYL. Management: Consider alternatives to this combination when possible. If the combination is used, monitor more closely for bradycardia, apnea, and excessive CNS depression. Propofol induction dose requirements may be reduced. Pediatric patients may be at greater risk. *Risk D: Consider Therapy Modification*

Fexinidazole: Propofol may increase arrhythmogenic effects of Fexinidazole. Propofol may increase QTc-prolonging effects of Fexinidazole. *Risk X: Avoid*

Finerenone: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Finerenone. *Risk C: Monitor*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flibanserin: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fluorouracil Products: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Gadobenate Dimeglumine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Gadobenate Dimeglumine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Halofantrine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Halofantrine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Haloperidol: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Haloperidol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

HydroXYZine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Inotuzumab Ozogamicin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Inotuzumab Ozogamicin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Ixabepilone. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Lemborexant. Management: The maximum recommended dosage of lemborexant is 5 mg, no more than once per night, when coadministered with weak CYP3A4 inhibitors. *Risk D: Consider Therapy Modification*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levoketoconazole: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Lofexidine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Lomitapide: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Lomitapide. Management: Decrease lomitapide dose 50% if starting a weak CYP3A4 inhibitor. Lomitapide dose may be up-titrated, but the max recommended dose is lower - 10 mg for patients aged 2 to 10 yrs; 20 mg for patients aged 11 to 17 yrs; 30 mg for patients 18 yrs and older. *Risk D: Consider Therapy Modification*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Meglumine Antimoniate: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Melatonin: May increase therapeutic effects of Propofol. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Methylergonovine: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Methylergonovine. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Midazolam: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Midazolam. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Midostaurin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Midostaurin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

NiMODipine: CYP3A4 Inhibitors (Weak) may increase serum concentrations of NiMODipine. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

OLANzapine: CNS depressant effects of OLANzapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ondansetron: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Ondansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

Oxytocin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Oxytocin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pentamidine (Systemic): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Pentamidine (Systemic). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pimozide: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Pimozide. *Risk X: Avoid*

Pimozide: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piperaquine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Probucol: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Probucol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): Propofol may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Antipsychotics (Moderate Risk): May increase QTc-prolonging effects of Propofol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Class IC Antiarrhythmics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk) may increase serum concentrations of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of Propofol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Remdesivir: May increase bradycardic effects of Propofol. Remdesivir may increase QTc-prolonging effects of Propofol. Management: Monitor for bradycardia, QTc interval prolongation, and ventricular arrhythmias when remdesivir and propofol are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

RifAMPin: May increase hypotensive effects of Propofol. Management: Avoid this combination if possible. Use of propofol in a patient who has been taking rifampin may result in clinically significant hypotension. *Risk D: Consider Therapy Modification*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Sebetralstat: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Sebetralstat. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Simvastatin: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Simvastatin. CYP3A4 Inhibitors (Weak) may increase active metabolite exposure of Simvastatin. *Risk C: Monitor*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Sirolimus (Conventional): CYP3A4 Inhibitors (Weak) may increase serum concentrations of Sirolimus (Conventional). *Risk C: Monitor*

Sirolimus (Protein Bound): CYP3A4 Inhibitors (Weak) may increase serum concentrations of Sirolimus (Protein Bound). Management: Reduce the dose of protein bound sirolimus to 56 mg/m² when used concomitantly with a weak CYP3A4 inhibitor. *Risk D: Consider Therapy Modification*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tacrolimus (Systemic): CYP3A4 Inhibitors (Weak) may increase serum concentrations of Tacrolimus (Systemic). *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Terbutaline: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these

agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thioridazine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Toremifene: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Toremifene. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Triazolam: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Triazolam. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Tripolidine: CNS Depressants may increase CNS depressant effects of Tripolidine. *Risk C: Monitor*

Ubrogepant: CYP3A4 Inhibitors (Weak) may increase serum concentrations of Ubrogepant. Management: In patients taking weak CYP3A4 inhibitors, the initial and second dose (given at least 2 hours later if needed) of ubrogepant should be limited to 50 mg. *Risk D: Consider Therapy Modification*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: May increase therapeutic effects of Propofol. Management: Decrease the propofol dose and monitor for sedation and cardiorespiratory depression during coadministration with valproate products. *Risk D: Consider Therapy Modification*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Food Interactions

Edetate disodium, an ingredient of propofol emulsion, may lead to decreased zinc levels in patients on prolonged therapy (>5 days) or those predisposed to deficiency (burns, diarrhea, and/or major sepsis). Management: Zinc replacement therapy may be needed.

Pregnancy Considerations

Propofol crosses the placenta and may be associated with neonatal CNS and respiratory depression. Monitor exposed neonates for hypotonia and sedation.

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D-aspartate (NMDA) receptors and/or potentiate gamma-aminobutyric acid (GABA) activity may affect brain development. Evaluate benefits and potential risks of fetal exposure to propofol when duration of surgery is expected to be >3 hours (Olutoye 2018).

Propofol has been used as an induction agent in cases where general anesthesia is needed for cesarean delivery (ACOG 209 2019; Devroe 2015).

The ACOG recommends that pregnant patients should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

Propofol is present in breast milk.

In studies where propofol is used prior to cesarean delivery, exposure to the infant is low due to low concentrations in breast milk and low volume of breast milk produced within 24 hours' postpartum. When measurable, concentrations decrease quickly over time (Dailland 1989). Studies conducted in nine lactating women undergoing general surgery where propofol was used to induce anesthesia report propofol milk concentrations that are <1% of the total maternal dose. Except in one case where age was not clearly specified, infants in these studies were between 6 weeks and 15 months of age (Nitsun 2006; Stuttmann 2010).

A green discoloration to the breast milk was noted in a woman following administration of propofol during surgery for removal of an ectopic pregnancy. Although other medications were also administered, propofol was detected in the milk and assumed to be the cause; resolution of this effect occurred within 48 hours after surgery (Birkholz 2009). Dizziness or drowsiness were not observed in four breastfed infants. One was a 3-month old infant fed as early as ~90 minutes after maternal surgery (Stuttmann 2010).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk

supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Dietary Considerations

Propofol is formulated in an oil-in-water emulsion. If on parenteral nutrition, may need to adjust the amount of lipid infused. Propofol emulsion contains 1.1 kcal/mL. Soybean fat emulsion is used as a vehicle for propofol. Formulations also contain egg phospholipids (egg lecithin) and glycerol.

Monitoring Parameters

Cardiac monitor, BP, oxygen saturation (during monitored anesthesia care sedation), arterial blood gas (with prolonged infusions). With prolonged infusions (eg, ICU sedation), monitor for signs and symptoms of propofol-related infusion syndrome: Metabolic acidosis, hyperkalemia, rhabdomyolysis or elevated CPK, hepatomegaly, and progression of cardiac and renal failure.

ICU sedation: Assess and adjust sedation according to scoring system (Richmond Agitation-Sedation Scale or Sedation-Agitation Scale) (SCCM [Devlin 2018]); assess CNS function daily. Serum triglyceride levels should be obtained prior to initiation of therapy and every 3 to 7 days thereafter, especially if receiving for >48 hours and/or doses ≥ 50 mcg/kg/minute (Devlin 2005); use IV port opposite propofol infusion or temporarily suspend infusion and flush port prior to blood draw.

Diprivan: Monitor zinc levels in patients predisposed to deficiency (burns, diarrhea, major sepsis) or after 5 days of treatment.

Mechanism of Action

Propofol is a short-acting, lipophilic intravenous general anesthetic. The drug is unrelated to any of the currently used barbiturate, opioid, benzodiazepine, arylcyclohexylamine, or imidazole intravenous anesthetic agents. Propofol causes global CNS depression, presumably through agonism of GABA_A receptors and perhaps reduced glutamatergic activity through NMDA receptor blockade.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Anesthetic: Bolus infusion (dose dependent): 9 to 51 seconds (average 30 seconds)

Duration: 3 to 10 minutes depending on the dose, rate and duration of administration; with prolonged use (eg, 10 days ICU sedation), propofol accumulates in tissues and redistributes into plasma when the drug is discontinued, so that the time to awakening (duration of action) is increased; however, if dose is titrated on a daily basis, so that the minimum effective dose is utilized, time to awakening may be within 10 to 15 minutes even after prolonged use

Distribution: Large volume of distribution; highly lipophilic ; V_d :

Children 4 to 12 years: 5 to 10 L/kg

Adults: 2 to 10 L/kg; after a 10-day infusion, V_d approaches 60 L/kg; decreased in the elderly

Protein binding: 97% to 99%

Metabolism: Hepatic to water-soluble sulfate and glucuronide conjugates (~50%)

Half-life elimination: Biphasic: Initial: 40 minutes; Terminal: 4 to 7 hours (after 10-day infusion, may be up to 1 to 3 days)

Excretion: Urine (~88% as metabolites, 40% as glucuronide metabolite); feces (<2%)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: With increasing age, the dose requirement decreases because of occurrence of higher peak plasma concentrations.

Therapeutic hypothermia: Hypothermia (33°C to 34°C or 91.4°F to 93.2°F) is associated with a decrease in total body clearance (~23%) and an increase in plasma concentrations (~28%), likely as

a result of reduced hepatic blood flow and/or reduced intercompartmental clearance (Bjelland 2013; Leslie 1995). During the rewarming phase, plasma concentrations decrease (~12% or ~3% decrease for every centigrade increase in core body temperature from 33°C to 37°C [91.4°F to 98.6°F]) (Bjelland 2014).

Patient Counseling Points

(For additional information see "Propofol: Patient drug information")

What is this drug used for?

- It is used for anesthesia before or during some procedures or surgery.
- It is used to cause sleep while on a ventilator.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Feeling sleepy
- Trouble sleeping
- Cough after waking up

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Pancreas problem (pancreatitis) like very bad stomach pain, very bad back pain, or very bad upset stomach or throwing up
- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Trouble breathing, slow breathing, or shallow breathing
- Fast or slow heartbeat
- Trouble controlling body movements, twitching, change in balance, trouble swallowing or speaking
- A burning, numbness, or tingling feeling that is not normal
- Propofol infusion syndrome (PRIS) like feeling confused, very sleepy, or very tired or weak; dark urine or not able to pass urine; fast breathing; a fast or abnormal heartbeat; muscle pain or weakness; severe stomach pain, upset stomach, or throwing up; or shortness of breath, a big weight gain, or swelling in the arms or legs
- Injection site reaction like redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to

provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Diprivan | Propofol lipuro | Provive;
- (AR) Argentina: Diprivan | Fresofol mct | Gobbifol | Oleo lax | Profol | Propofol Abbott | Propofol Baxter | Propofol gray | Propofol lipuro | Propofol northia | Propovan;
- (AT) Austria: Diprivan | Propofol Baxter | Propofol claris | Propofol lipuro | Propofol ratiopharm;
- (AU) Australia: Diprivan | Fresofol mct/lct | Propofol bc | Propofol lipuro | Provive | Provive mct lct | Recofol;
- (BD) Bangladesh: Pofol;
- (BE) Belgium: Diprivan | Propofol EG | Propofol fresenius | Propofol sandoz;
- (BG) Bulgaria: Diprivan | Dormofol | Plofed | Profol | Propofol sandoz;
- (BR) Brazil: Diprivan | Fresofol | Hyfol | Lipuro | Mayoprofol | Oponap | Profolen | Pronest | Propoab-bott | Propotil | Propovan | Provive | Spiva mct lct;
- (CH) Switzerland: Disoprivan | Propofol lipuro | Propofol mct fresenius;
- (CL) Chile: Diprivan | Prosafe;
- (CN) China: Di shi ning | Propofol fresenius | Recofol;
- (CO) Colombia: Anecorta | Blofop | Diprivan | Fresofol | Mayoprofol | Nirfol | Oleo lax | Profol | Rapifol | Serafol | Sevofol | Spiva mct lct | Sysnof | Troypofol;
- (CR) Costa Rica: Diprivan | Oleo lax | Propofol gray;
- (CZ) Czech Republic: Diprivan | Propofol lipuro | Propofol mct fresenius | Recofol;
- (DE) Germany: Anesia | Diprivan | Disoprivan | Klimofol | Propofol claris | Propofol fresenius | Propofol Hexal | Propofol lipuro | Propofol ratiopharm | Recofol;
- (DO) Dominican Republic: Diprivan | Propovan | Proxofol;
- (EC) Ecuador: Diprivan | Fiprolif | Pofol | Profol | Propofol lipuro | Propovan | Provive | Troypofol;
- (EE) Estonia: Anesia | Diprivan | Propofol claris | Propofol lipuro | Propofol noramedia | Propoven;
- (EG) Egypt: Diprivan | Pofol | Propofol lipuro | Recofol;
- (ES) Spain: Diprivan | Ivofol | Propofol Abbott | Propofol Baxter | Propofol fresenius | Propofol lipomed | Propofol lipoven fresenius | Propofol lipuro | Propofol rovi | Propofol sandoz | Recofol;
- (ET) Ethiopia: Spiva;
- (FI) Finland: Diprivan | Profast | Propofol Abbott | Propofol Alpharma | Propofol fresenius kabi | Propofol lipuro | Propolipid | Recofol;
- (FR) France: Diprivan | Propofol Baxter | Propofol bioq | Propofol dakota | Propofol fresenius | Propofol lipuro | Propofol panpharma | Propofol pf;
- (GB) United Kingdom: Diprivan | Propofol lipuro | Propoven;
- (GR) Greece: Diprivan | Propofol lipuro | Propofol mct/lct fresenius | Propofol/Alvia | Propofol/baxter | Propofol/fresenius | Recofol | Spiva;
- (HK) Hong Kong: Diprivan | Propofol lipuro;
- (HR) Croatia: Propofol fresenius | Propofol lipuro;
- (HU) Hungary: Anesia | Diprivan | Propofol fresenius | Propofol lipuro | Propofol mct/lct fresenius | Recofol;
- (ID) Indonesia: Anesticap | Diprivan | Fresofol | Nupovel | Proanes | Propofol lipuro | Recofol | Recofol n | Safol | Trivam;
- (IE) Ireland: Diprivan | Propofol lipuro | Propoven;
- (IN) India: Cleofol | Critifol | Diprivan | Epuro | Fresofol | Fresofol mct/lct | Hetrofol | Hyprovan | Neorof | Popoglan | Profol | Propovan | Prosthetik | Rapifol | Rofol | Troypofol;
- (IT) Italy: Diprivan | Propofol b.braun | Propofol kabi | Rapiva | Ripol | Unifol;
- (JO) Jordan: Diprivan | Safol;

- (JP) Japan: Diprivan | Fresofol | Propofol Abbott | Propofol fk | Propofol Fuji | Propofol Hospira | Propofol Maruishi | Propofol Merck;
- (KE) Kenya: Diprivan | Dormifor | Galaxy's propofol | Neorof | Pofol | Propofol lipuro | Propovan | Provive;
- (KR) Korea, Republic of: Anepol | Aquafol | Diprivan | Fresofol | Pofol | Propofol lipuro | Provive;
- (KW) Kuwait: Diprivan | Fresenius propoven | Propofol lipuro | Recofol;
- (LB) Lebanon: Diprivan | Dormifor;
- (LT) Lithuania: Anesia | Diprivan | Propofol claris | Propofol lipuro | Propofol norameda | Propoven | Provive;
- (LU) Luxembourg: Diprivan;
- (LV) Latvia: Anesia | Diprivan | Propofol claris | Propofol lipuro | Propoven;
- (MA) Morocco: Diprivan;
- (MX) Mexico: Braunolax | Diovist | Diprivan | Fresofol | Indufol | Lipronest | Lofol | Moveloc | Neleprit | Piraloxdil | Plofed | Prigadef | Quipro | Quishvan | Rachechi | Recofol | Vanxiprol | Vitenaltec | Xendta | Xenprof;
- (MY) Malaysia: Diprivan | Fresofol | Fresofol mct/lct | Propofol lipuro | Spiva | Troypofol;
- (NG) Nigeria: Anesia | Propofol lipuro | Troypofol;
- (NL) Netherlands: Diprivan | Propofol farmaconetwork | Propofol fresenius | Propofol Genthon | Propofol lipuro | Propofol mct/lct fresenius;
- (NO) Norway: Diprivan | Profast | Propofol braun | Propofol lipuro | Propolipid | Recofol;
- (NZ) New Zealand: Diprivan | Fresofol | Provive | Provive mct lct | Recofol;
- (PA) Panama: Diprivan | Pofol | Profol | Propofol usp;
- (PE) Peru: Diprivan | Fresofol | Nirfol | Propovan | Prosafe | Troypofol;
- (PH) Philippines: Fresofol | Iv pro | Lexofol | Lipuro | Neorof | Operol | Profogen | Spiva with mct lct;
- (PK) Pakistan: Fresofol | Pofol | Propofol lipuro;
- (PL) Poland: Abbofol | Anesia | Diprivan | Disoprivan | Plofed | Propofol Baxter | Propofol fresenius | Propofol lipuro | Propofol mct/lct fresenius | Propofol norameda | Propofol sandoz | Provive | Troypofol;
- (PR) Puerto Rico: Diprivan | Fresenius propoven | Propoven;
- (PT) Portugal: Propofol Alparma | Propofol aps | Propofol Genthon | Propofol lipuro | Propofol norameda | Recofol;
- (PY) Paraguay: Diprivan | Dorminox | Propofol claris;
- (QA) Qatar: Diprivan | Propofol (Lipuro) | Propofol 1% MCT/LCT Fresenius | Propofol-Lipuro | Provive | Recofol EDTA | Recofol N | Respofol | Sayfol;
- (RO) Romania: Diprivan | Profast | Propofol mct/lct fresenius;
- (RU) Russian Federation: Diprivan | Pofol | Proanes | Propofol kabi | Propofol lipuro | Propofol medargo | Propofol novo | Propovan | Provive | Recofol;
- (SA) Saudi Arabia: Diprivan | Propoven | Recofol;
- (SE) Sweden: Diprivan | Profast | Propofol Abbott | Propofol Actavis | Propofol fresenius | Propofol lipuro | Propofol sandoz | Propolipid | Recofol;
- (SG) Singapore: Diprivan | Fresofol | Propofol lipuro | Recofol;
- (SI) Slovenia: Anesia | Diprivan | Propofol lipuro | Propoven;
- (SK) Slovakia: Diprivan | Propofol lipuro | Propofol mct/lct fresenius | Recofol;
- (TH) Thailand: Anepol | Anesvan | Diprivan | Fresofol | Fresofol mct/lct | Pofol | Propofol Abbott | Propofol lipuro | Recofol | Troypofol;
- (TN) Tunisia: Pms propofol | Propofol fresenius | Propofol lipuro;
- (TR) Turkey: Diprivan | Dormofol | Pofol | Propofol Abbott | Propofol Genthon | Propofol pf | Propofol pf mct;
- (TW) Taiwan: Anesvan | Diprivan | Diprofen | Fresofol | Pofol | Propofol lipuro | Recofol;
- (UA) Ukraine: Diprivan | Diprofol | Diprofol edta | Profol | Propofol kabi | Propofol lipuro | Propofol rompharm | Propophol | Recofol | Rekofol;
- (UG) Uganda: Fresenius propoven | Hyprovan | Neorof | Proanes mct/lct | Provive;
- (UY) Uruguay: Diprivan | Ivofol | Oleo lax | Pofol | Profolen | Propofol mct/lct fresenius | Propovan;
- (VE) Venezuela, Bolivarian Republic of: Anespro | Diprivan | Diprivan aventis | Profol | Profol spiva | Propowell | Spiva;

- (ZA) South Africa: B. braun propofol | Diprivan | Propoven | Provive;
- (ZM) Zambia: Troypofol;
- (ZW) Zimbabwe: Fresenius propoven

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REFERENCES

1. Ahlfors CE. Benzyl alcohol, kernicterus, and unbound bilirubin. *J Pediatr.* 2001;139(2):317-319. [PubMed 11487763]
2. American Academy of Allergy, Asthma, and Immunology. Soy-allergic and egg-allergic patients can safely receive anesthesia. Updated September 28, 2020. Accessed August 16, 2021. <https://www.aaaai.org/Tools-for-the-Public/Conditions-Library/Allergies/soy-egg-anesthesia>
3. American College of Obstetricians and Gynecologists (ACOG). ACOG committee opinion no. 775: nonobstetric surgery during pregnancy. *Obstet Gynecol.* 2019;133(4):e285-e286. [PubMed 30913200]
4. American College of Obstetricians and Gynecologists (ACOG). ACOG practice bulletin no. 209: obstetric analgesia and anesthesia. *Obstet Gynecol.* 2019;133(3):e208-e225. [PubMed 30801474]
5. Amorós A, Aparicio JR, Garmendia M, Casellas JA, Martínez J, Jover R. Deep sedation with propofol does not precipitate hepatic encephalopathy during elective upper endoscopy. *Gastrointest Endosc.* 2009;70(2):262-268. doi:10.1016/j.gie.2008.10.038 [PubMed 19394004]
6. Andolfatto G, Abu-Laban RB, Zed PJ, et al. Ketamine-propofol combination (ketofol) versus propofol alone for emergency department procedural sedation and analgesia: a randomized double-blind trial. *Ann Emerg Med.* 2012;59(6):504-12.e1-2. doi:10.1016/j.annemergmed.2012.01.017 [PubMed 22401952]
7. Araújo AM, Machado H, de Pinho PG, Soares-da-Silva P, Falcão A. Population pharmacokinetic-pharmacodynamic modeling for propofol anesthesia guided by the bispectral index (BIS). *J Clin Pharmacol.* 2019. doi:10.1002/jcph.1560 [PubMed 31797395]
8. Araújo AM, Machado HS, Falcão AC, Soares-da-Silva P. Reliability of body-weight scalars on the assessment of propofol induction dose in obese patients. *Acta Anaesthesiol Scand.* 2018;62(4):464-473. doi:10.1111/aas.13039 [PubMed 29159892]
9. Asghar MU, Cheema HA, Tanveer K, Leinwand J. Propofol infusion and acute pancreatitis: a review. *Am J Ther.* 2020;27(4):e371-e374. doi:10.1097/MJT.0000000000001021 [PubMed 31283535]
10. Asserhøj LL, Mosbech H, Krøigaard M, Garvey LH. No evidence for contraindications to the use of propofol in adults allergic to egg, soy or peanut. *Br J Anaesth.* 2016;116(1):77-82. [PubMed 26675952]
11. Bagley L, Kordun A, Sinnott S, Lobo K, Cravero J. Food allergy history and reaction to propofol administration in a large pediatric population. *Paediatr Anaesth.* 2021;31(5):570-577. doi:10.1111/pan.14147 [PubMed 33529424]
12. Barr J, Fraser GL, Puntillo K, et al; American College of Critical Care Medicine. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med.* 2013;41(1):263-306. doi:10.1097/CCM.0b013e3182783b72 [PubMed 23269131]
13. Based on expert opinion.
14. Bassi E, Ferreira CB, Macedo E, Malbouisson LM. Recurrent clotting of dialysis filter associated with hypertriglyceridemia induced by propofol. *Am J Kidney Dis.* 2014;63(5):860-861. doi:10.1053/j.ajkd.2014.01.417 [PubMed 24529993]

15. Baumeister FAM, Oberhoffer R, Liebhaber GM, et.al. Fatal propofol infusion syndrome in association with ketogenic diet. *Neuropediatrics*. 2004;35: 250-252. [PubMed 15328567]
16. Baumgarten A, Newton M. Propofol-induced uric acid crystals: a case of cloudy urine. *Paediatr Anaesth*. 2021;31(10):1132-1133. doi:10.1111/pan.14260 [PubMed 34309948]
17. Bedirli N, Egirtas O, Cosarcin K, Bozkirli F. A comparison of fentanyl with tramadol during propofol-based deep sedation for pediatric upper endoscopy. *Paediatr Anaesth*. 2012;22(2):150-155. [PubMed 21958025]
18. Belso N, Kui R, Szegesdi I, et al. Propofol and fentanyl induced perioperative anaphylaxis. *Br J Anaesth*. 2011;106(2):283-284. doi:10.1093/bja/aeq384 [PubMed 21233125]
19. Bendel S, Ruokonen E, Pölönen P, Uusaro A. Propofol causes more hypotension than etomidate in patients with severe aortic stenosis: a double-blind, randomized study comparing propofol and etomidate. *Acta Anaesthesiol Scand*. 2007;51(3):284-289. doi:10.1111/j.1399-6576.2006.01206.x [PubMed 17390417]
20. Benken S, Madrzyk E, Chen D, et al. Hemodynamic effects of propofol and dexmedetomidine in septic patients without shock. *Ann Pharmacother*. 2020;54(6):533-540. doi:10.1177/1060028019895502 [PubMed 31849243]
21. Birkholz T, Eckardt G, Renner S, et al. Green breast milk after propofol administration. *Anesthesiology*. 2009;111(5):1168-1169. [PubMed 19858894]
22. Bjelland TW, Klepstad P, Haugen BO, Nilsen T, Dale O. Effects of hypothermia on the disposition of morphine, midazolam, fentanyl, and propofol in intensive care unit patients. *Drug Metab Dispos*. 2013;41(1):214-223. doi:10.1124/dmd.112.045567 [PubMed 23115086]
23. Bjelland TW, Klepstad P, Haugen BO, Nilsen T, Salvesen O, Dale O. Concentrations of remifentanyl, propofol, fentanyl, and midazolam during rewarming from therapeutic hypothermia. *Acta Anaesthesiol Scand*. 2014;58(6):709-715. doi:10.1111/aas.12300 [PubMed 24611449]
24. Brophy GM, Bell R, Claassen J, et al; Neurocritical Care Society Status Epilepticus Guideline Writing Committee. Guidelines for the evaluation and management of status epilepticus. *Neurocrit Care*. 2012;17(1):3-23. [PubMed 22528274]
25. Caro D. Induction agents for rapid sequence intubation in adults for emergency medicine and critical care. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 31, 2020.
26. Centers for Disease Control (CDC). Neonatal deaths associated with use of benzyl alcohol—United States. *MMWR Morb Mortal Wkly Rep*. 1982;31(22):290-291. <http://www.cdc.gov/mmwr/preview/mmwr> [PubMed 6810084]
27. Chidambaran V, Costandi A, D'Mello A. Correction to: Propofol: a review of its role in pediatric anesthesia and sedation. *CNS Drugs*. 2018;32(9):873. doi:10.1007/s40263-018-0561-1 [PubMed 30101390]
28. Chidambaran V, Costandi A, D'Mello A. Propofol: a review of its role in pediatric anesthesia and sedation [published correction appears in *CNS Drugs*. 2018;32(9):873]. *CNS Drugs*. 2015;29(7):543-563. doi:10.1007/s40263-015-0259-6 [PubMed 26290263]
29. Chukwuemeka A, Ko R, Ralph-Edwards A. Short-term low-dose propofol anaesthesia associated with severe metabolic acidosis. *Anaesth Intensive Care*. 2006;34(5):651-655. doi:10.1177/0310057X0603400503 [PubMed 17061643]
30. Corbett SM, Montoya ID, Moore FA. Propofol-related infusion syndrome in intensive care patients. *Pharmacotherapy*. 2008;28(2):250-258. doi:10.1592/phco.28.2.250 [PubMed 18225970]
31. Corrado MJ, Kovacevic MP, Dube KM, Lupi KE, Szumita PM, DeGrado JR. The incidence of propofol-induced hypertriglyceridemia and identification of associated risk factors. *Crit Care Explor*. 2020;2(12):e0282. doi:10.1097/CCE.0000000000000282 [PubMed 33274340]

32. Cravero JP, Kaplan RF, Landrigan-Ossar M, Coté CJ. Sedation for diagnostic and therapeutic procedures outside the operating room. In: Coté CJ, Lerman J, Anderson B, eds. *A Practice of Anesthesia for Infants and Children*. 6th ed. Elsevier; 2019:chap. 48.
33. Dailland P, Cockshott ID, Lirzin JD, et al. Intravenous propofol during cesarean section: placental transfer, concentrations in breast milk, and neonatal effects. A preliminary study. *Anesthesiology*. 1989;71(6):827-834. [PubMed 2589672]
34. de Kort EHM, Prins SA, Reiss IKM, et al. Propofol for endotracheal intubation in neonates: a dose-finding trial. *Arch Dis Child Fetal Neonatal Ed*. 2020a;105(5):489-495. doi:10.1136/archdischild-2019-318474 [PubMed 31932363]
35. de Kort EHM, Twisk JWR, van T Verlaat EPG, Reiss IKM, Simons SHP, van Weissenbruch MM. Propofol in neonates causes a dose-dependent profound and protracted decrease in blood pressure. *Acta Paediatr*. 2020b;109(12):2539-2546. doi:10.1111/apa.15282 [PubMed 32248549]
36. de Leon-Casasola OA, Weiss A, Lema MJ. Anaphylaxis due to propofol. *Anesthesiology*. 1992;77(2):384-386. doi:10.1097/0000542-199208000-00026 [PubMed 1642359]
37. de Wit F, van Vliet AL, de Wilde RB, et al. The effect of propofol on haemodynamics: cardiac output, venous return, mean systemic filling pressure, and vascular resistances. *Br J Anaesth*. 2016;116(6):784-789. doi:10.1093/bja/aew126 [PubMed 27199311]
38. Deutschman CS, Harris AP, Fleisher LA. Changes in heart rate variability under propofol anesthesia: a possible explanation for propofol-induced bradycardia. *Anesth Analg*. 1994;79(2):373-377. doi:10.1213/00000539-199408000-00031 [PubMed 7639382]
39. Devlin JW, Lau AK, Tanios MA. Propofol-associated hypertriglyceridemia and pancreatitis in the intensive care unit: an analysis of frequency and risk factors. *Pharmacotherapy*. 2005;25(10):1348-1352. doi:10.1592/phco.2005.25.10.1348 [PubMed 16185179]
40. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018;46(9):e825-e873. doi: 10.1097/CCM.0000000000003299 [PubMed 30113379]
41. Devroe S, Van de Velde M, Rex S. General anesthesia for caesarean section. *Curr Opin Anaesthesiol*. 2015;28(3):240-246. [PubMed 25827280]
42. Diedrich DA, Brown DR. Analytic reviews: propofol infusion syndrome in the ICU. *J Intensive Care Med*. 2011;26(2):59-72. doi:10.1177/0885066610384195 [PubMed 21464061]
43. Dietrich SK, Mixon MA, Rogoszewski RJ, et al. Hemodynamic effects of propofol for induction of rapid sequence intubation in traumatically injured patients. *Am Surg*. 2018;84(9):1504-1508. [PubMed 30268185]
44. Diprivan (propofol) [prescribing information]. Lake Zurich, IL: Fresenius Kabi; February 2025.
45. Diprivan (propofol) [product monograph]. Oakville, Ontario, Canada: Aspen Pharmacare Canada Inc; July 2025.
46. Douglas RJ, Cadogan M. Cardiac arrhythmia during propofol sedation. *Emerg Med Australas*. 2008;20(5):437-440. doi:10.1111/j.1742-6723.2008.01126.x [PubMed 18973642]
47. Drislane FW. Convulsive status epilepticus in adults: Management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 31, 2020.
48. Ducart AR, Watremez C, Louagie YA, Collard EL, Broka SM, Joucken KL. Propofol-induced anaphylactoid reaction during anesthesia for cardiac surgery. *J Cardiothorac Vasc Anesth*. 2000;14(2):200-201. doi:10.1016/s1053-0770(00)90020-x [PubMed 10794344]
49. Dziedzic A. Is Propofol Safe For Food Allergy Patients? A Review of the Evidence. *SAAD Dig*. 2016;32:23-27. [PubMed 27145556]

50. Ebrahimi F, Semela D, Heim M. Impact of propofol sedation on the diagnostic accuracy of hepatic venous pressure gradient measurements in patients with cirrhosis. *Hepatol Int.* 2022;16(4):817-823. doi:10.1007/s12072-021-10261-z [PubMed 34699037]
51. Engstrom K, Brown CS, Mattson AE, Lyons N, Rech MA. Pharmacotherapy optimization for rapid sequence intubation in the emergency department. *Am J Emerg Med.* 2023;70:19-29. doi:10.1016/j.ajem.2023.05.004 [PubMed 37196592]
52. Erdil F, Demirbilek S, Begec Z, Ozturk E, Ersoy MO. Effects of propofol or etomidate on QT interval during electroconvulsive therapy. *J ECT.* 2009;25(3):174-177. [PubMed 19225403]
53. Erstad BL, Barletta JF. Drug dosing in the critically ill obese patient-a focus on sedation, analgesia, and delirium. *Crit Care.* 2020;24(1):315. doi:10.1186/s13054-020-03040-z [PubMed 32513237]
54. Expert opinion. Senior Hepatic Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; SasanSakiani, MD.
55. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
56. Farzana N, Tewari P, Sureka S, Dixit A. Parosmia and dysgeusia after intravenous propofol-based general anesthesia: a case report. *Ann Card Anaesth.* 2022;25(1):112-115. doi:10.4103/aca.ACA_93_21 [PubMed 35075034]
57. Fassoulaki A, Farinotti R, Servin F, et al, "Chronic Alcoholism Increases the Induction Dose of Propofol in Humans," *Anesth Analg*, 1993, 77(3):553-6. [PubMed 8368555]
58. Ferguson I, Bell A, Treston G, New L, Ding M, Holdgate A. Propofol or ketofol for procedural sedation and analgesia in emergency medicine-the POKER study: a randomized double-blind clinical trial. *Ann Emerg Med.* 2016;68(5):574-582.e1. doi:10.1016/j.annemergmed.2016.05.024 [PubMed 27460905]
59. Fong JJ, Sylvia L, Ruthazer R, Schumaker G, Kcomt M, Devlin JW. Predictors of mortality in patients with suspected propofol infusion syndrome. *Crit Care Med.* 2008;36(8):2281-2287. doi:10.1097/CCM.0b013e318180c1eb [PubMed 18664783]
60. Fung BM, Leon DJ, Beck LN, Tabibian JH. Pre-procedural preparation and sedation for gastrointestinal endoscopy in patients with advanced liver disease. *Dig Dis Sci.* 2022;67(7):2739-2753. doi:10.1007/s10620-021-07111-1 [PubMed 34169430]
61. Gangineni K, Scase AE, and Fearn J, "Propofol and Peanut Allergy," *Anaesthesia*, 2007, 62(11):1183-92. [PubMed 17924915]
62. Godwin SA, Burton JH, Gerardo CJ, et al. Clinical policy: procedural sedation and analgesia in the emergency department. *Ann Emerg Med.* 2014;63(2):247-258. doi: 10.1016/j.annemergmed.2013.10.015 [PubMed 24438649]
63. Guacho JAL, de Moura DTH, Ribeiro IB, et al. Propofol vs midazolam sedation for elective endoscopy in patients with cirrhosis: a systematic review and meta-analysis of randomized controlled trials. *World J Gastrointest Endosc.* 2020;12(8):241-255. doi:10.4253/wjge.v12.i8.241 [PubMed 32879659]
64. Hanna JP, Ramundo ML. Rhabdomyolysis and hypoxia associated with prolonged propofol infusion in children. *Neurology.* 1998;50(1):301-303. doi:10.1212/wnl.50.1.301 [PubMed 9443502]
65. Hemphill S, McMenamin L, Bellamy MC, Hopkins PM. Propofol infusion syndrome: a structured literature review and analysis of published case reports. *Br J Anaesth.* 2019;122(4):448-459. doi:10.1016/j.bja.2018.12.025 [PubMed 30857601]
66. Hermann R, Vettermann J. Change of ectopic supraventricular tachycardia to sinus rhythm during administration of propofol. *Anesth Analg.* 1992;75(6):1030-1032. doi:10.1213/00000539-199212000-00028 [PubMed 1443685]

67. Hidaka S, Kawamoto M, Kurita S, Yuge O. Comparison of the effects of propofol and midazolam on the cardiovascular autonomic nervous system during combined spinal and epidural anesthesia. *J Clin Anesth*. 2005;17(1):36-43. doi:10.1016/j.jclinane.2004.03.012 [PubMed 15721728]
68. Hughes CG, Mailloux PT, Devlin JW, et al. Dexmedetomidine or propofol for sedation in mechanically ventilated adults with sepsis. *N Engl J Med*. 2021;384(15):1424-1436. doi:10.1056/NEJMoa2024922 [PubMed 33528922]
69. Hume-Smith HV, Sanatani S, Lim J, Chau A, Whyte SD. The effect of propofol concentration on dispersion of myocardial repolarization in children. *Anesth Analg*. 2008;107(3):806-810. [PubMed 18713888]
70. "Inactive" ingredients in pharmaceutical products: update (subject review). American Academy of Pediatrics (AAP) Committee on Drugs. *Pediatrics*. 1997;99(2):268-278. [PubMed 9024461]
71. Jakob SM, Ruokonen E, Grounds RM, et al; Dexmedetomidine for Long-Term Sedation Investigators. Dexmedetomidine vs midazolam or propofol for sedation during prolonged mechanical ventilation: two randomized controlled trials. *JAMA*. 2012;307(11):1151-1160. doi:10.1001/jama.2012.304 [PubMed 22436955]
72. Janmahasatian S, Duffull SB, Ash S, Ward LC, Byrne NM, Green B. Quantification of lean bodyweight. *Clin Pharmacokinet*. 2005;44(10):1051-1065. doi:10.2165/00003088-200544100-00004 [PubMed 16176118]
73. Johnson JL, Hawthorne A, Bounds M, Weldon DJ. New perspectives on propofol allergy. *Am J Health Syst Pharm*. 2021;78(24):2195-2203. doi:10.1093/ajhp/zxab298 [PubMed 34309659]
74. Jones GM, Doepker BA, Erdman MJ, Kimmons LA, Eliovich L. Predictors of severe hypotension in neurocritical care patients sedated with propofol. *Neurocrit Care*. 2014;20(2):270-276. doi:10.1007/s12028-013-9902-6 [PubMed 24233892]
75. Kim DH, Kweon TD, Nam SB, Han DW, Cho WY, Lee JS. Effects of target concentration infusion of propofol and tracheal intubation on QTc interval. *Anaesthesia*. 2008;63(10):1061-1064. [PubMed 18717665]
76. Koul A, Jain R, Sood J. A critical incident report: propofol triggered anaphylaxis. *Indian J Anaesth*. 2011;55(5):530-533. doi:10.4103/0019-5049.89898 [PubMed 22174476]
77. Krajčová A, Waldauf P, Anděl M, Duška F. Propofol infusion syndrome: a structured review of experimental studies and 153 published case reports. *Crit Care*. 2015;19:398. doi:10.1186/s13054-015-1112-5 [PubMed 26558513]
78. Krauss B, Green SM. Procedural sedation and analgesia in children. *Lancet*. 2006;367(9512):766-780. [PubMed 16517277]
79. Kress JP, Pohlman AS, O'Connor MF, Hall JB. Daily interruption of sedative infusions in critically ill patients undergoing mechanical ventilation. *N Engl J Med*. 2000;342(20):1471-1477. doi:10.1056/NEJM200005183422002 [PubMed 10816184]
80. Kumar AN, Schwartz DE, Lim KG. Propofol-induced pancreatitis: recurrence of pancreatitis after rechallenge. *Chest*. 1999;115(4):1198-1199. doi:10.1378/chest.115.4.1198 [PubMed 10208230]
81. Laxenaire MC, Mata-Bermejo E, Moneret-Vautrin DA, Gueant JL. Life-threatening anaphylactoid reactions to propofol (Diprivan). *Anesthesiology*. 1992;77(2):275-280. doi:10.1097/00000542-199208000-00009 [PubMed 1379418]
82. Laxenaire MC, Mertes PM; Groupe d'Etudes des Réactions Anaphylactoïdes Peranesthésiques. Anaphylaxis during anaesthesia. Results of a two-year survey in France. *Br J Anaesth*. 2001;87(4):549-558. doi:10.1093/bja/87.4.549 [PubMed 11878723]
83. Lee SJ, Kim SI, Jung BI, et al. Suspected anaphylactic reaction associated with microemulsion propofol during anesthesia induction. *J Korean Med Sci*. 2012;27(7):827-829. doi:10.3346/jkms.2012.27.7.827 [PubMed 22787384]

84. Legriel S, Oddo M, Brophy GM. What's new in refractory status epilepticus? *Intensive Care Med.* 2017;43(4):543-546. doi:10.1007/s00134-016-4501-6 [PubMed 27544138]
85. Leslie K, Sessler DI, Bjorksten AR, Moayeri A. Mild hypothermia alters propofol pharmacokinetics and increases the duration of action of atracurium. *Anesth Analg.* 1995;80(5):1007-1014. doi:10.1097/00000539-199505000-00027 [PubMed 7726398]
86. Lieberman P, Nicklas RA, Randolph C, et al. Anaphylaxis--a practice parameter update 2015. *Ann Allergy Asthma Immunol.* 2015;115(5):341-384. [PubMed 26505932]
87. Liu Q, Kong AL, Chen R, et al. Propofol and arrhythmias: two sides of the coin. *Acta Pharmacol Sin.* 2011;32(6):817-823. doi:10.1038/aps.2011.42 [PubMed 21642950]
88. McConachie I, Keaveny JP, Healy TE, Vohra S, Million L. Effect of anaesthesia on the QT interval. *Br J Anaesth.* 1989;63(5):558-560. [PubMed 2605073]
89. McHale SP, Konieczko K. Anaphylactoid reaction to propofol. *Anaesthesia.* 1992;47(10):864-865. doi:10.1111/j.1365-2044.1992.tb03150.x [PubMed 1443479]
90. McLaughlin DC, Fang DC, Nolot BA, Guru PK. Hypertriglyceridemia causing continuous renal replacement therapy dysfunction in a patient with end-stage liver disease. *Indian J Nephrol.* 2018;28(4):303-306. doi:10.4103/ijn.IJN_201_17 [PubMed 30158750]
91. McLeay SC, Morrish GA, Kirkpatrick CM, Green B. Encouraging the move towards predictive population models for the obese using propofol as a motivating example. *Pharm Res.* 2009;26(7):1626-1634. doi:10.1007/s11095-009-9873-7 [PubMed 19337821]
92. McNeill O, Kerridge RK, Boyle MJ. Review of procedures for investigation of anaesthesia-associated anaphylaxis in Newcastle, Australia. *Anaesth Intensive Care.* 2008;36(2):201-207. doi:10.1177/0310057X0803600210 [PubMed 18361011]
93. McPherson C, Meyers RS, Thackray J, et al. Pediatric Pharmacy Association 2025 KIDs List of key potentially inappropriate drugs in pediatrics. *J Pediatr Pharmacol Ther.* 2025;30(4):422-439. doi:10.5863/JPPT-25-00061 [PubMed 40821415]
94. Miller KA, Andolfatto G, Miner JR, Burton JH, Krauss BS. Clinical practice guideline for emergency department procedural sedation with propofol: 2018 update. *Ann Emerg Med.* 2019;73(5):470-480. doi:10.1016/j.annemergmed.2018.12.012 [PubMed 30732981]
95. Mirrakhimov AE, Voore P, Halytskyy O, Khan M, Ali AM. Propofol infusion syndrome in adults: a clinical update. *Crit Care Res Pract.* 2015;2015:260385. doi:10.1155/2015/260385 [PubMed 25954513]
96. Mosier JM, Sakles JC. The physiologically difficult airway: optimization of airway management in adults at risk of decompensation for emergency medicine and critical care. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed May 9, 2025.
97. Mudri M, Williams A, Priestap F, Davidson J, Merritt N. Comparison of drugs used for intubation of pediatric trauma patients. *J Pediatr Surg.* 2020;55(5):926-929. doi:10.1016/j.jpedsurg.2020.01.041 [PubMed 32067810]
98. Muniraj T, Aslanian HR. Hypertriglyceridemia independent propofol-induced pancreatitis. *JOP.* 2012;13(4):451-453. doi:10.6092/1590-8577/822 [PubMed 22797405]
99. Murphy A, Campbell DE, Baines D, et al. Allergic reactions to propofol in egg-allergic children. *Anesth Analg.* 2011;113(1):140-144. [PubMed 21467558]
100. Nishiyama T, Hanaoka K. Propofol-induced bronchoconstriction: two case reports. *Anesth Analg.* 2001;93(3):645-646. doi:10.1097/00000539-200109000-00022 [PubMed 11524333]
101. Nitsun M, Szokol JW, Saleh HJ, et al. Pharmacokinetics of midazolam, propofol, and fentanyl transfer to human breast milk. *Clin Pharmacol Ther.* 2006;79(6):549-557. [PubMed 16765143]

102. Olutoye OA, Baker BW, Belfort MA, Olutoye OO. Food and Drug Administration warning on anesthesia and brain development: implications for obstetric and fetal surgery. *Am J Obstet Gynecol*. 2018;218(1):98-102. [PubMed 28888583]
103. Ong J, Van Gerpen R. Recommendations for management of noncytotoxic vesicant extravasations. *J Infus Nurs*. 2020;43(6):319-343. doi:10.1097/NAN.0000000000000392 [PubMed 33141794]
104. Page RL 2nd, O'Bryant CL, Cheng D, et al; American Heart Association Clinical Pharmacology and Heart Failure and Transplantation Committees of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Drugs That May Cause or Exacerbate Heart Failure: A Scientific Statement From the American Heart Association [published correction appears in *Circulation*. 2016;134(12):e261]. *Circulation*. 2016;134(6):e32-e69. [PubMed 27400984]
105. Panchamia JK, Amundson AW, Demirci O, Braksick SA, Wijdicks EF. A case report of "propofol frenzy": anesthetic considerations for a severe propofol-induced neuroexcitation reaction. *A A Pract*. 2022;16(3):e01569. doi:10.1213/XAA.0000000000001569 [PubMed 35299226]
106. Parness J, Savard M, Turgeon AF. Propofol-related infusion syndrome heralding a mitochondrial disease: case report. *Neurology*. 2014;82(5):461. doi:10.1212/01.wnl.0000443933.18269.20 [PubMed 24491974]
107. Prasad A, Worrall BB, Bertram EH, Bleck TP. Propofol and midazolam in the treatment of refractory status epilepticus. *Epilepsia*. 2001;42(3):380-386. [PubMed 11442156]
108. Rai S, Drislane FW. Treatment of refractory and super-refractory status epilepticus. *Neurotherapeutics*. 2018;15(3):697-712. doi:10.1007/s13311-018-0640-5 [PubMed 29922905]
109. Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine. ABM clinical protocol #15: analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med*. 2017;12(9):500-506. [PubMed 29624435]
110. Refer to the manufacturer's labeling.
111. Riezzo I, Centini F, Neri M, et al. Brugada-like EKG pattern and myocardial effects in a chronic propofol abuser. *Clin Toxicol (Phila)*. 2009;47(4):358-363. doi:10.1080/15563650902887842 [PubMed 19514884]
112. Roberts J. *Roberts and Hedges' Clinical Procedures in Emergency Medicine and Acute Care*. 7th ed. Elsevier; 2019.
113. Roberts RJ, Barletta JF, Fong JJ, et al. Incidence of propofol-related infusion syndrome in critically ill adults: a prospective, multicenter study. *Crit Care*. 2009;13(5):R169. doi:10.1186/cc8145 [PubMed 19874582]
114. Robinson BJ, Ebert TJ, O'Brien TJ, Colinto MD, Muzi M. Mechanisms whereby propofol mediates peripheral vasodilation in humans. Sympathoinhibition or direct vascular relaxation? *Anesthesiology*. 1997;86(1):64-72. doi:10.1097/00000542-199701000-00010 [PubMed 9009941]
115. Roden DM. Drug-induced prolongation of the QT interval. *N Engl J Med*. 2004;350(10):1013-1022. doi:10.1056/NEJMra032426 [PubMed 14999113]
116. Rosero. Monitored anesthesia care in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 21, 2020.
117. Saarnivaara L, Hiller A, Oikonen M. QT interval, heart rate and arterial pressures using propofol, thiopentone or methohexitone for induction of anaesthesia in children. *Acta Anaesthesiol Scand*. 1993;37(4):419-423. [PubMed 8322572]
118. Saarnivaara L, Klemola UM, Lindgren L, Rautiainen P, Suvanto A. QT interval of the ECG, heart rate and arterial pressure using propofol, methohexital or midazolam for induction of anaesthesia. *Acta Anaesthesiol Scand*. 1990;34(4):276-281. [PubMed 2343728]

119. Sakabe M, Fujiki A, Inoue H. Propofol induced marked prolongation of QT interval in a patient with acute myocardial infarction. *Anesthesiology*. 2002;97(1):265-266. [PubMed 12131130]
120. Samal S, Singhanian N, Adl D, Afzal O, Singhanian G. Chocolate milk in CRRT circuit: propofol induced hypertriglyceridemia. *Am J Med Sci*. 2021;361(1):135-136. doi:10.1016/j.amjms.2020.08.003 [PubMed 32950175]
121. Savard M, Dupré N, Turgeon AF, Desbiens R, Langevin S, Brunet D. Propofol-related infusion syndrome heralding a mitochondrial disease: case report. *Neurology*. 2013;81(8):770-771. doi:10.1212/WNL.0b013e3182a1aa78 [PubMed 23873972]
122. Servin F, Cockshott ID, Farinotti R, Haberer JP, Winckler C, Desmonts JM. Pharmacokinetics of propofol infusions in patients with cirrhosis. *Br J Anaesth*. 1990;65(2):177-183. doi:10.1093/bja/65.2.177 [PubMed 2223333]
123. Servin F, Desmonts JM, Farinotti R, Haberer JP, Winckler C. Pharmacokinetics of propofol administered by continuous infusion in patients with cirrhosis. Preliminary results. *Anaesthesia*. 1988a;43 suppl:23-24. doi:10.1111/j.1365-2044.1988.tb09063.x [PubMed 3259091]
124. Servin F, Desmonts JM, Haberer JP, Cockshott ID, Plummer GF, Farinotti R. Pharmacokinetics and protein binding of propofol in patients with cirrhosis. *Anesthesiology*. 1988b;69(6):887-891. doi:10.1097/00000542-198812000-00014 [PubMed 3264120]
125. Shearin AE, Patanwala AE, Tang A, Erstad BL. Predictors of hypotension associated with propofol in trauma patients. *J Trauma Nurs*. 2014;21(1):4-8. doi:10.1097/JTN.0000000000000022 [PubMed 24399312]
126. Simons SH, van der Lee R, Reiss IK, van Weissenbruch MM. Clinical evaluation of propofol as sedative for endotracheal intubation in neonates. *Acta Paediatr*. 2013;102(11):e487-492. [PubMed 23889264]
127. Smith HAB, Besunder JB, Betters KA, et al. 2022 Society of Critical Care Medicine (SCCM) clinical practice guidelines on prevention and management of pain, agitation, neuromuscular blockade, and delirium in critically ill pediatric patients with consideration of the ICU environment and early mobility. *Pediatr Crit Care Med*. 2022;23(2):e74-e110. doi:10.1097/PCC.00000000000002873 [PubMed 35119438]
128. Sommerfield DL, Lucas M, Schilling A, et al. Propofol use in children with allergies to egg, peanut, soybean or other legumes. *Anaesthesia*. 2019;74(10):1252-1259. doi:10.1111/anae.14693 [PubMed 31077340]
129. Stefanos SS, Kiser TH, MacLaren R, Mueller SW, Reynolds PM. Management of noncytotoxic extravasation injuries: a focused update on medications, treatment strategies, and peripheral administration of vasopressors and hypertonic saline. *Pharmacotherapy*. 2023;43(4):321-337. doi:10.1002/phar.2794 [PubMed 36938775]
130. Stollings JL, Diedrich DA, Oyen LJ, Brown DR. Rapid-sequence intubation: a review of the process and considerations when choosing medications. *Ann Pharmacother*. 2014;48(1):62-76. doi:10.1177/1060028013510488 [PubMed 24259635]
131. Stuttmann R, Schäfer C, Hilbert P, Meyer MR, Maurer HH. The breast feeding mother and xenon anaesthesia: four case reports. Breast feeding and xenon anaesthesia. *BMC Anesthesiol*. 2010;10:1 [PubMed 20167123]
132. Tam MK, Irwin MG, Tse ML, Lui YW, Law KI, Ng PW. Prolonged myoclonus after a single bolus dose of propofol [published correction appears in *Anaesthesia*. 2010;65(2):222]. *Anaesthesia*. 2009;64(11):1254-1257. doi:10.1111/j.1365-2044.2009.06035.x [PubMed 19825064]
133. Tanskanen PE, Kytä JV, Randell TT. QT interval and QT dispersion during the induction of anaesthesia in patients with subarachnoid haemorrhage: a comparison of thiopental and propofol. *Eur J Anaesthesiol*. 2002;19(10):749-754. [PubMed 12463387]

134. Tisdale JE, Chung MK, Campbell KB, et al. Drug-induced arrhythmias: a scientific statement from the American Heart Association. *Circulation*. 2020;142(15):e214-e233. doi:10.1161/CIR.0000000000000900 [PubMed 32929996]
135. Tisdale JE, Jaynes HA, Kingery JR, et al. Development and validation of a risk score to predict QT interval prolongation in hospitalized patients [published correction appears in *Circ Cardiovasc Qual Outcomes*. 2013;6(6):e57]. *Circ Cardiovasc Qual Outcomes*. 2013;6(4):479-487. doi:10.1161/CIRCOUTCOMES.113.000152 [PubMed 23716032]
136. Tramèr MR, Moore RA, McQuay HJ. Propofol and bradycardia: causation, frequency and severity. *Br J Anaesth*. 1997;78(6):642-651. doi:10.1093/bja/78.6.642 [PubMed 9215013]
137. US Food and Drug Administration. *FDA Drug Safety Communication: FDA review results in new warnings about using general anesthetics and sedation drugs in young children and pregnant women*. US Food and Drug Administration Web site. <https://www.fda.gov/Drugs/DrugSafety/ucm554634.htm>. Published December 14, 2016. Updated April 27, 2017. Accessed May 26, 2017.
138. Vuillaume C, Geeraerts T, Rollin A, Fourcade O, Maury P. ST elevation, long QT, and ventricular arrhythmias related to propofol infusion syndrome without biological abnormalities in a child after head injury. *The Journal of Innovations in Cardiac Rhythm Management*. 2014;5:1508-1511. doi:10.19102/icrm.2014.050105
139. Vossler DG, Bainbridge JL, Boggs JG, et al. Treatment of refractory convulsive status epilepticus: a comprehensive review by the American Epilepsy Society Treatments Committee. *Epilepsy Curr*. Published online August 21, 2020. doi:10.1177/1535759720928269 [PubMed 32822230]
140. Waheed MA, Oud L. Acute pulmonary edema associated with propofol: an unusual complication. *West J Emerg Med*. 2014;15(7):845-848. doi:10.5811/westjem.2014.7.22942 [PubMed 25493132]
141. Warpechowski P, dos Santos AT, Pereira PJ, de Lima GG. Effects of propofol on the cardiac conduction system. *Rev Bras Anesthesiol*. 2010;60(4):438-444. doi:10.1016/S0034-7094(10)70054-4 [PubMed 20659617]
142. Welzing L, Kribs A, Eifinger F, Huenseler C, Oberthuer A, Roth B. Propofol as an induction agent for endotracheal intubation can cause significant arterial hypotension in preterm neonates. *Paediatr Anaesth*. 2010;20(7):605-611. [PubMed 20642659]
143. Wiskin AE, Smith J, Wan SK, Nally MW, Shah N. Propofol anaesthesia is safe in children with food allergy undergoing endoscopy. *Br J Anaesth*. 2015;115(1):145-146. doi:10.1093/bja/aev177 [PubMed 26089466]
144. Witenko CJ, Littlefield AJ, Abedian S, An A, Barie PS, Berger K. The safety of continuous infusion propofol in mechanically ventilated adults with coronavirus disease 2019. *Ann Pharmacother*. Published online May 14, 2021. doi:10.1177/10600280211017315 [PubMed 33985368]
145. Xu G, Qiao N, Pan Y, Simayi A, Chen N. The appropriate dose of propofol for anesthesia induction in morbidly obese patients. *Ann Palliat Med*. 2020;9(4):1921-1927. doi:10.21037/apm-20-1223 [PubMed 32692244]
146. You BC, Jang AS, Han JS, et al. A case of propofol-induced oropharyngeal angioedema and bronchospasm. *Allergy Asthma Immunol Res*. 2012;4(1):46-48. doi:10.4168/aaair.2012.4.1.46 [PubMed 22211170]
147. Zaki HA, Shallik N, Shaban E, et al. An analytical comparison between ketamine alone and a combination of ketamine and propofol (ketofol) for procedural sedation and analgesia from an emergency perspective: a systematic review and meta-analysis. *Cureus*. 2022;14(7):e27318. doi:10.7759/cureus.27318 [PubMed 36042988]
148. Zarovnyaya EL, Jobst BC, Harris BT. Propofol-associated fatal myocardial failure and rhabdomyolysis in an adult with status epilepticus. *Epilepsia*. 2007;48(5):1002-1006. doi:10.1111/j.1528-1167.2007.01042.x [PubMed 17381434]

149. Zhou W, Fontenot HJ, Liu S, Kennedy RH. Modulation of cardiac calcium channels by propofol. *Anesthesiology*. 1997;86(3):670-675. doi:10.1097/00000542-199703000-00020 [PubMed 9066334]

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